Aalok Aryal

Electrical Engineer

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PROFILE

Electrical Engineer with academic research experience in fault diagnosis and predictive maintenance of induction motors. Skilled in MATLAB/Simulink, AutoCAD and Python; applied experience in machine learning and embedded systems. Seeking roles in electrical design, power systems and industrial automation.

EDUCATION

Bachelor of Electrical Engineering — **2022 – 2026**
Pulchowk Campus, IOE, Tribhuvan University · , Lalitpur, Nepal — *Aggregrate: 85%*

Relevant Coursework: Power System Analysis, Electrical Machines, Power Electronics, Control System, Switchgear & Protection, High Voltage Engineering, Signal Analysis, Electrical Machine Design

Higher Secondary Education – Science & Mathematics — **2019 – 2021**
St. Xavier's College · Maitighar, Kathmandu — *Cumulative Grade: A+*

KEY PROJECTS

Fault Diagnosis of Induction Motors — **2026**
MATLAB, Simulink, Python — *Major Project – Curriculum*

- Modeled broken rotor bar faults in the dq0 reference frame under varying load conditions; generated synthetic training data with empirically calibrated noise injection to approximate real-world signal characteristics.
- Identified poor generalization of ML models trained purely on simulation data to real-world stator current signals; addressed the sim-to-real gap using a CNN-LSTM transfer learning framework.
- Achieved 99% classification accuracy and F1-score on the IEEE benchmark dataset using a novel CNN-LSTM transfer learning framework, outperforming conventional simulation-trained models.
- Paper presented in the Symposium EPNC-2026 in Tallinn, Estonia in June

Finite Element Modeling of Induction Machine — **2026**
COMSOL Multiphysics

- Modeled and simulated induction machine electromagnetic performance using FEM to analyze flux distribution, torque characteristics, and losses under various load conditions.

Electrical System Design of a Hospital Building — **2025**
AutoCAD, DiaLUX

- Designed lighting layouts in DiaLUX ensuring adequate illuminance across clinical, corridor, and utility spaces; produced detailed AutoCAD drawings for light circuits, power circuits, and the distribution system of a six-floor hospital building (four above-ground, two basement). with standard lighting design requirements.
- Sized and selected cables and protective devices including MCBs, and MCCBs; designed distribution board configuration with circuit breaker coordination across all floors.

Power System Load Flow Analysis — **2024**
MATLAB

- Implemented Gauss-Seidel Method, Newton-Raphson Method and Fast Decoupled Method of load flow analysis via MATLAB programs; analyzed convergence, computational efficiency and numerical stability

Interactive Pong Game on ESP32 — **2024**
C/C++, ESP32, TFT Display

- Developed a real-time single and two-player Pong game on an ESP32 microcontroller, rendering graphics on a TFT display via SPI communication with score tracking and potentiometer-based paddle control.

TECHNICAL SKILLS

Programming Languages:	C, C++, Python, MATLAB
Engineering Software:	MATLAB/Simulink, COMSOL Multiphysics, AutoCAD, DiaLUX
Documentation:	Microsoft Office Suite, LaTeX
Soft Skills:	Team Collaboration, Problem Solving, Attention to Detail

CERTIFICATIONS

Supervised Machine Learning: Regression and Classification **Apr 2025**
DeepLearning.AI & Stanford Online, Coursera

MATLAB Onramp & Simulink Onramp **2024**
MathWorks

LANGUAGES & INTERESTS

Languages:	Nepali (Native), English (Fluent), Hindi (Conversational), Spanish (Basic)
Interests:	Power Systems, Renewable Energy, Embedded Systems, Machine Learning

REFERENCES

Asst. Prof. Bishal Silwal
Department of Electrical Engineering, Pulchowk Campus, IoE *bishal.silwal@pcampus.edu.np*

Asst. Prof. Anil Kumar Panjiyar
Department of Electrical Engineering, Pulchowk Campus, IoE *anil.panjiyar@pcampus.edu.np*